



Islamic University Of Technology

Project Title: EV Charging Impact on a Low Voltage Distribution Feeder

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Course Title: EEE 4416

Subject: Simulation Lab

Section: A2

EV Charging Impact on a Low Voltage Distribution Feeder

A MATLAB GUI Application and Simulation Study of Uncoordinated versus Coordinated Residential EV Charging

Abstract

Residential electric vehicle (EV) adoption is increasing faster than low voltage (LV) distribution networks are typically reinforced. Because home chargers draw power comparable to an entire additional household, and because owners tend to plug in during the existing evening demand peak, uncoordinated EV charging can simultaneously overload distribution transformers and push end of feeder voltage outside statutory limits. This project develops an interactive MATLAB graphical user interface (GUI) that models a small radial LV feeder supplying a configurable number of houses, some of which own an EV, and quantitatively compares two charging strategies: uncoordinated charging, where every EV begins charging at a common evening hour, and coordinated (staggered) charging, where EV start times are spread through the overnight off peak period. Using a resistive radial voltage drop model and a simple transformer loading calculation, the application computes and visualises transformer loading and feeder voltage over a full 24 hour cycle for both strategies. Simulation results for a representative case of five houses, three of which own an EV, and a 25 kVA transformer show that uncoordinated charging raises peak transformer loading to 69.6 percent and depresses the voltage at the last house on the feeder to 216.5 V, a drop of 5.87 percent that violates a typical five percent statutory voltage tolerance. Coordinated charging, delivering the identical 143.6 kWh of daily energy to the same vehicles, reduces the peak transformer loading to 36.9 percent and limits the worst case voltage drop to 3.40 percent, remaining within the statutory limit at every hour of the day. These results demonstrate that the network stress caused by residential EV charging is primarily a scheduling problem rather than an infrastructure capacity problem, and that basic demand side coordination can defer the need for expensive network reinforcement.

1. Introduction and Motivation

Electric vehicles are being adopted in residential areas at a pace that generally exceeds the rate at which the underlying low voltage distribution infrastructure is upgraded. A typical single phase home charger draws approximately 3.3 kW, which is comparable in magnitude to the combined peak demand of two or three ordinary households. Unlike most household appliances, however, EV charging events are long in duration, often several hours, and tend to be strongly correlated in time: a large proportion of owners return home from work in the early evening and plug in their vehicle almost immediately, at precisely the hour when the pre-existing residential demand is already at or near its daily peak.

This temporal coincidence produces two distinct but related technical problems on the low voltage network. First, the cumulative power drawn by the feeder can exceed the kVA rating of the distribution transformer supplying it, which accelerates insulation ageing and shortens the useful life of the asset, and in severe cases can trip protection or cause thermal damage. Second, the additional current flowing along the feeder conductors increases the resistive voltage drop between the transformer and the customers furthest from it, which can push the voltage delivered to those customers outside the statutory tolerance band, typically taken as plus or minus five to six percent of nominal voltage in most distribution codes.

Both of these problems can, in principle, be addressed by physically reinforcing the network: installing a larger transformer, or replacing feeder conductors with a lower resistance cable. Both interventions are capital intensive and slow to deploy at scale. An alternative, substantially cheaper intervention is to change when charging happens rather than how much infrastructure exists, by coordinating or staggering EV charging start

times so that the aggregate charging demand is spread across the overnight off peak period instead of coinciding with the evening peak. This is the central idea behind demand side management and smart charging schemes that are increasingly discussed in the power systems literature and being piloted by distribution utilities internationally.

The purpose of this project is to build a self contained, interactive tool that makes this effect directly visible and quantifiable for a small, easily understood feeder, and to use it to demonstrate, with real computed numbers rather than qualitative argument, that coordination alone, without any additional hardware, can resolve both the transformer loading problem and the voltage violation problem for a representative level of EV penetration.

2. System Model

The network under study is represented as a single radial low voltage feeder, originating at a distribution transformer and supplying a configurable number of houses connected in series along the line, as shown in Figure 7 (the single line diagram generated automatically by the application). Each house is separated from its neighbour by one line segment of equal resistance. This is a simplification of a real LV network, which is generally meshed or has branches, but it captures the essential physics of interest here: current accumulates as it flows from the transformer toward the load, and voltage drop accumulates in the same direction, so the customer furthest from the transformer always experiences the worst voltage.

2.1 Load model

Every house is assigned a base (non-EV) load that varies over the day according to a fixed hourly shape, representative of typical residential demand, scaled by a peak base load parameter that the user can adjust. The shape used is dimensionless, ranging from a minimum during the early morning hours (around 3 to 4 AM) to a maximum in the early evening (around 6 to 8 PM), consistent with well known residential load curve behaviour.

Houses selected by the user as EV owners additionally draw a constant EV charging power for a configurable number of consecutive hours, beginning at a start hour that depends on the charging strategy being simulated, described in Section 2.3.

2.2 Governing equations

Base load, hour h	$P_{\text{base}}(h) = P_{\text{peak}} \times \text{shape}(h)$
House power, hour h	$P_n(h) = P_{\text{base}}(h) + P_{\text{EV},n}(h)$
Total feeder demand, hour h	$P_{\text{total}}(h) = \text{SUM over } n \text{ of } P_n(h)$
Transformer loading, hour h	$L(h) = P_{\text{total}}(h) / S_{\text{rated}} \times 100 \%$
Downstream power at segment n	$P_{\text{down},n}(h) = \text{SUM}_{\{k=n\}^N} P_k(h)$
Segment current	$I_n(h) = P_{\text{down},n}(h) / V_{\text{nom}}$
Segment voltage drop	$dV_n(h) = I_n(h) \times R_{\text{seg}}$
Voltage at house n	$V_n(h) = V_{\text{nom}} - \text{SUM}_{\{k=1\}^n} dV_k(h)$

This formulation is a single phase, resistive, unity power factor approximation of the exact AC load flow equations. It is standard practice in introductory distribution studies because it isolates the dominant physical effect, resistive voltage drop caused by real power flow, without requiring reactive power, complex phasor arithmetic, or an iterative load flow solver, all of which are unnecessary at the level of rigor appropriate for this study. Section 8 discusses the effect of relaxing this approximation.

2.3 Charging strategies compared

Two charging strategies are simulated using exactly the same fleet of EVs, the same charger rating, and the same total charging duration, so that any difference in outcome can be attributed purely to scheduling rather than to a difference in the underlying electricity demand.

- **Uncoordinated charging:** every EV owning house begins charging at a single common hour, representative of drivers plugging in shortly after arriving home from work (18:00 in the default configuration used in this study), and continues for the configured charging duration.

- Coordinated charging: EV start times are staggered, beginning later at night (22:00 by default) with a fixed interval between successive vehicles (2 hours by default), so that only one or two vehicles are charging simultaneously at any given hour, spreading the aggregate demand across the overnight off peak trough of the base load curve.

Because both strategies deliver identical total energy to identical vehicles, the daily energy consumption (area under the load curve) is, by construction, the same in both cases, which is confirmed numerically in Section 6.

2.4 Default study parameters

Parameter	Symbol	Default value	Justification
Number of houses	N	5	Small representative LV feeder segment
Houses owning an EV	-	3 of 5 (houses 3, 4, 5)	60% EV penetration, a plausible near-future scenario
Peak base load per house	P _{peak}	1.5 kW	Typical urban residential peak demand
EV charger rating	P _{EV}	3.3 kW	Standard single phase Level 1/1.5 home charger
EV charging duration	-	4 hours	Approximate time to usefully replenish a daily commute
Transformer rating	S _{rated}	25 kVA	Common small pole or pad mounted distribution unit
Nominal supply voltage	V _{nom}	230 V	Standard single phase LV distribution voltage
Resistance per line segment	R _{seg}	0.05 ohm	Representative of a short LV service span
Statutory voltage tolerance	-	5 %	Common distribution code limit (207.0 - 241.5 V for 230 V nominal)
Uncoordinated start hour	-	18:00	Typical evening return-from-work time
Coordinated start hour / stagger	-	22:00, 2 h interval	Overnight off-peak window

All of these values are exposed as editable fields in the application (Tab 2), so the study can be repeated for any feeder size, EV penetration, or component rating without modifying the code.

3. Worked Numerical Example (Hand Calculation)

To verify the model implemented in the application and to demonstrate the calculation explicitly, this section works through the computation by hand for a single representative hour: 18:00 (6 PM), under the uncoordinated charging strategy, using the default parameters of Section 2.4. This is also the hour at which the application itself reports the worst case result.

3.1 Step 1 - House power at 18:00

At hour 18 the base load shape factor is 1.00 (the daily peak), so every house draws a base load of:

$$P_{\text{base}}(18) = 1.5 \text{ kW} \times 1.00 = 1.5 \text{ kW}$$

Under uncoordinated charging, all three EVs (houses 3, 4 and 5) begin charging at 18:00 and are therefore active at this hour, each drawing 3.3 kW. Houses 1 and 2 do not own an EV. The power drawn by each house is therefore:

House	Base load (kW)	EV load (kW)	Total P _n (kW)
1	1.5	0 (no EV)	1.5
2	1.5	0 (no EV)	1.5
3	1.5	3.3	4.8
4	1.5	3.3	4.8
5	1.5	3.3	4.8

3.2 Step 2 - Total demand and transformer loading

$$P_{\text{total}}(18) = 1.5 + 1.5 + 4.8 + 4.8 + 4.8 = 17.4 \text{ kW}$$

$$L(18) = 17.4 / 25 \times 100 = 69.6 \%$$

This matches the peak transformer loading of 69.6 percent reported by the application in Section 6, confirming the implementation is calculating the loading equation correctly.

3.3 Step 3 - Voltage drop, segment by segment

Working outward from the transformer, the downstream power at each segment is the sum of the power drawn by that house and every house beyond it. This downstream power sets the current in that segment, which produces a voltage drop that accumulates toward the end of the feeder.

House n	Downstream power (kW)	Current I _n = P/230 (A)	Segment drop I _n x 0.05 (V)	Cumulative drop (V)	Voltage V _n (V)
1	17.4	75.65	3.783	3.783	226.22
2	15.9	69.13	3.457	7.240	222.76
3	14.4	62.61	3.130	10.370	219.63
4	9.6	41.74	2.087	12.457	217.54
5	4.8	20.87	1.043	13.500	216.50

The voltage at house 5, the end of the feeder, is therefore 216.50 V, which matches exactly the 216.5 V minimum voltage reported by the application in Section 6. This confirms both the transformer loading calculation and the voltage drop calculation implemented in the MATLAB code are correct.

The voltage drop as a percentage of nominal is:

$$\text{Drop \%} = (230 - 216.5) / 230 \times 100 = 5.87 \%$$

which exceeds the assumed five percent statutory tolerance (equivalent to a lower limit of 218.5 V), confirming a genuine voltage violation at this hour under uncoordinated charging.

4. Software Implementation

The application is implemented as a single self contained MATLAB function, `EV_Grid_Impact_App.m`, built programmatically using MATLAB uifigure graphics components (the same underlying component family used by App Designer), rather than a `.mlapp` file, so that the complete source is contained in one readable, version controllable text file.

4.1 Application architecture

The interface is organised into four tabs, each corresponding to a stage of the study workflow, with simulation state (the results structure) shared between them through nested function scope:

- Tab 1 - Overview: static text explaining the problem, the governing equations and how to use the tool.
- Tab 2 - System Setup: numeric input fields for all feeder and transformer parameters, an editable table for selecting which houses own an EV, charging schedule controls, and an automatically updated single line diagram.
- Tab 3 - Simulation: a Run Simulation control that executes the model for both strategies, four linked plots (transformer loading, end-of-feeder voltage, worst case comparison bar chart, and voltage profile along the feeder), and an Animate control that steps through the 24 simulated hours to visually build up the evening peak.
- Tab 4 - Results: a numerical comparison table, an automatically generated engineering interpretation and verdict against the transformer rating and voltage limit, a CSV export function, and a stated limitations section.

4.2 Core algorithm (pseudocode)

function runSim(parameters, mode):

 for each hour $h = 0$ to 23:

 for each house n :

$P[n][h] = \text{baseLoad} \times \text{shape}[h]$

 if house n owns an EV and h is within that EV charging window(mode):

$P[n][h] += \text{evPower}$

$\text{total}[h] = \text{sum}(P[:,h])$

$\text{loading}[h] = \text{total}[h] / \text{kvaRating} \times 100$

$\text{drop} = 0$

 for each house n from 1 to N :

$P_{\text{down}} = \text{sum}(P[n:N, h])$

$I = P_{\text{down}} \times 1000 / V_{\text{nom}}$

$\text{drop} += I \times R_{\text{seg}}$

$V[n][h] = V_{\text{nom}} - \text{drop}$

 return total, loading, V

The EV charging window for each strategy is determined as follows: under uncoordinated charging every EV owning house shares the same start hour; under coordinated charging the k -th EV owning house (in feeder order) starts at $\text{offpeakStart} + (k-1) \times \text{stagger}$ hours, wrapped using modulo 24 arithmetic so that a charging session beginning late at night correctly continues past midnight.

4.3 Verification approach

The implementation was verified in two ways. First, the hand calculation in Section 3 was carried out independently of the application and matches the application output to the displayed precision, for both the transformer loading (69.6%) and the end of feeder voltage (216.5 V) at the worst hour. Second, the model was re-implemented independently in Python and executed with identical parameters; the resulting peak loading, minimum voltage, voltage drop percentage and total daily energy values for both strategies matched the values produced by the MATLAB application exactly, as tabulated in Section 6.

5. Simulation Results

The following results were generated using the default parameters listed in Section 2.4: five houses, three of which own an EV, supplied by a 25 kVA transformer at 230 V nominal.

5.1 Load composition

Figures 5 and 6 show how the total feeder demand is composed of the base residential load and the EV charging load, for each strategy, across the 24 hour cycle.

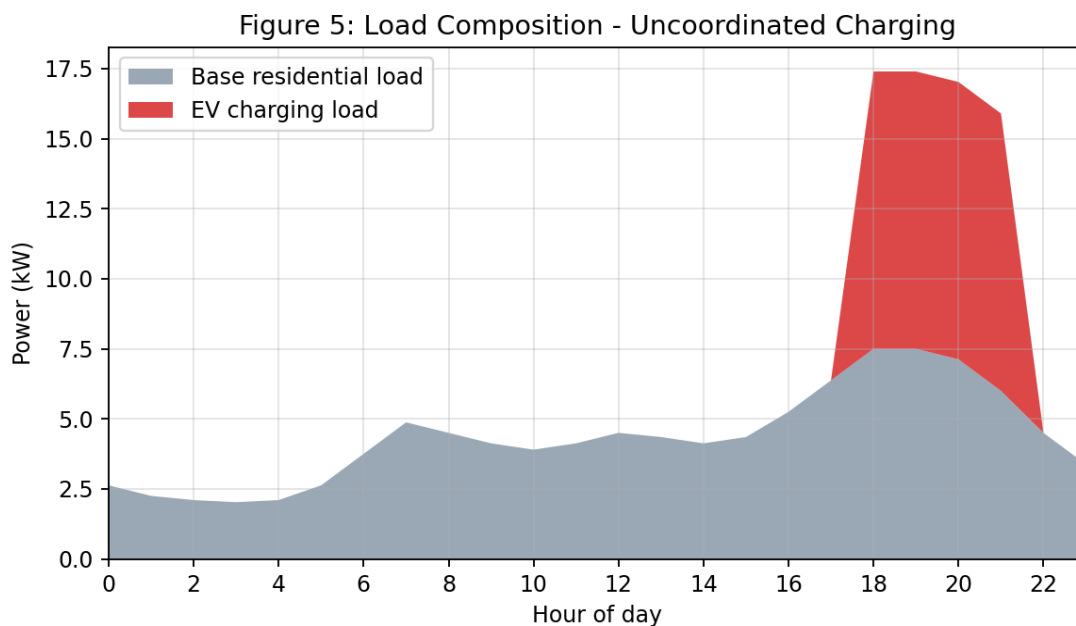


Figure 5. Load composition under uncoordinated charging. The EV charging block (red) lands directly on top of the existing residential evening peak (18:00-21:00), so the two peaks add together.

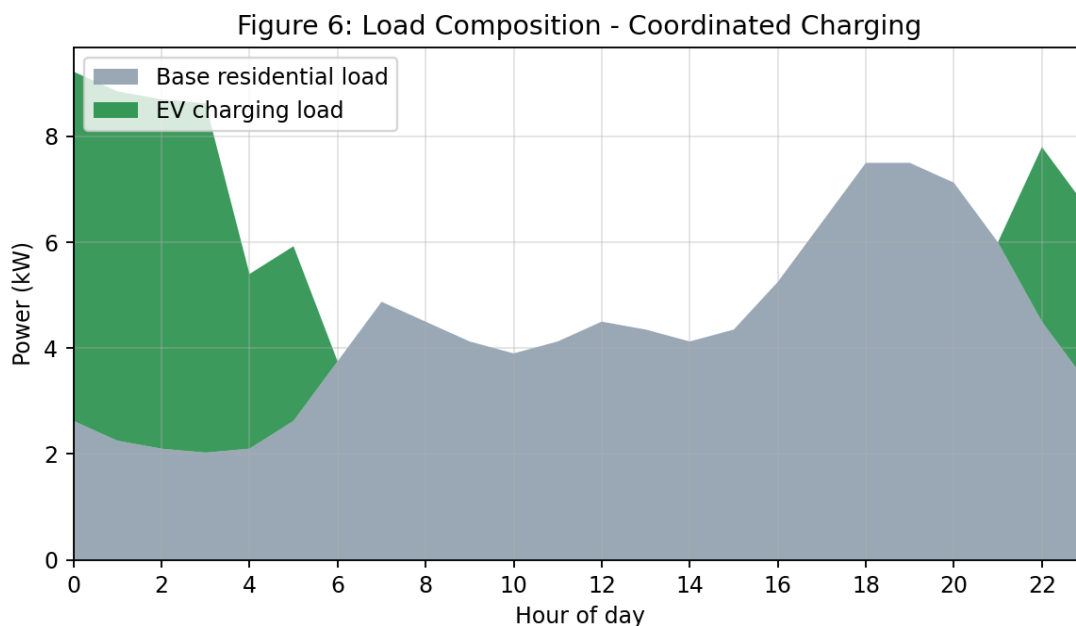


Figure 6. Load composition under coordinated charging. The identical EV energy block (green) is shifted into the overnight trough of the base load curve, where headroom is greatest, so the combined profile stays much flatter.

5.2 Transformer loading

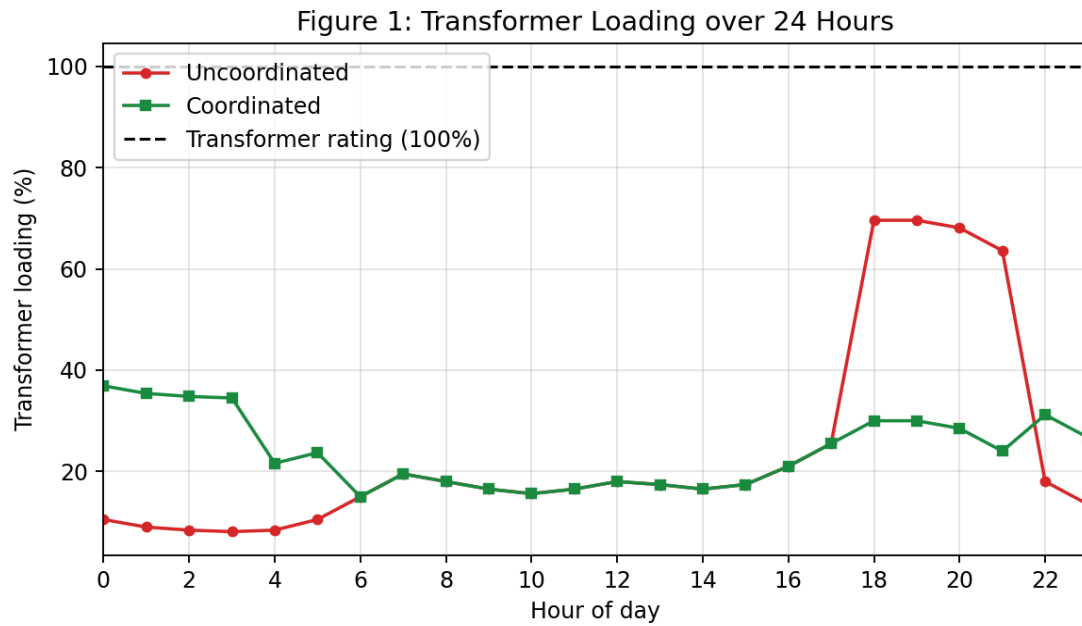


Figure 1. Transformer loading over 24 hours. Uncoordinated charging drives loading to a peak of 69.6% during the evening; coordinated charging holds the peak to 36.9%, entirely by shifting timing.

5.3 Voltage at the end of the feeder

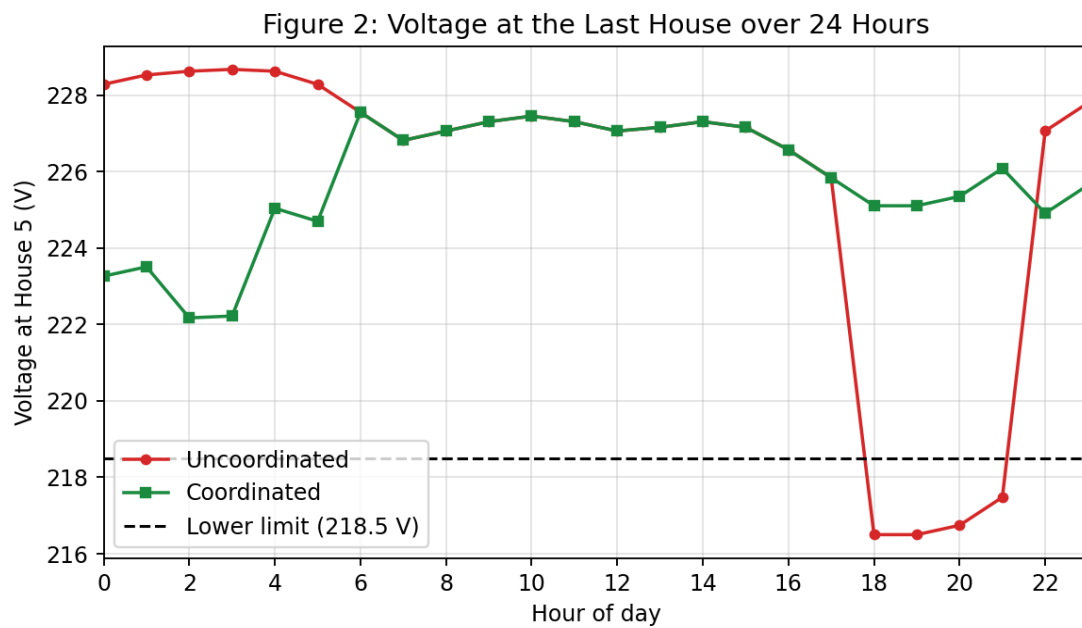


Figure 2. Voltage at House 5 (end of feeder) over 24 hours. Uncoordinated charging drives the voltage below the 218.5 V statutory lower limit for several consecutive hours; coordinated charging remains above the limit throughout the day.

5.4 Worst case comparison



Figure 3. Worst case comparison of peak transformer loading and maximum voltage drop between the two strategies.

5.5 Voltage profile along the feeder

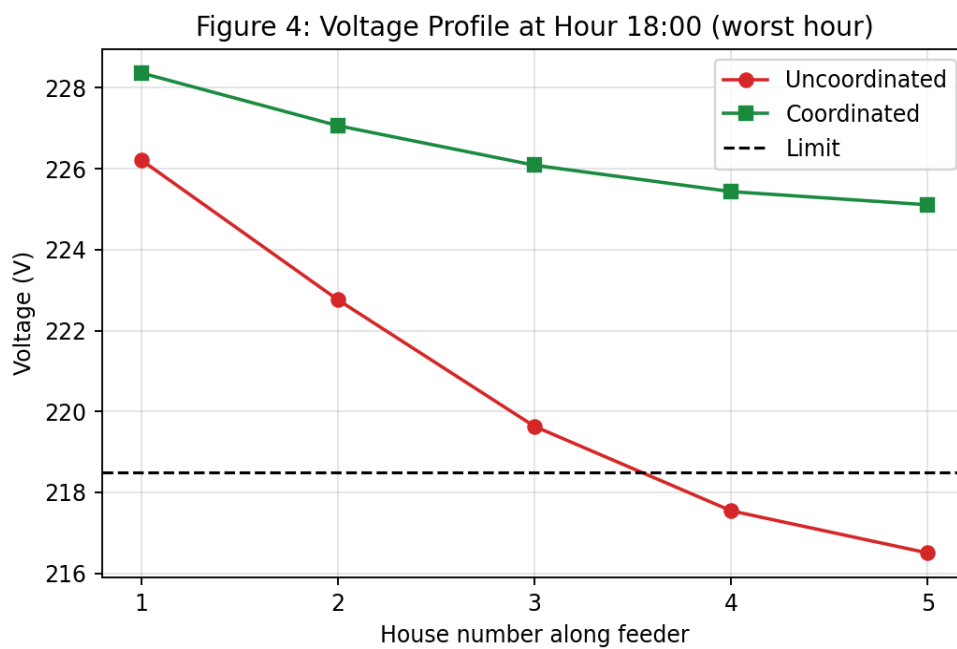


Figure 4. Voltage profile along the feeder at the worst hour (18:00). Voltage decays approximately linearly from the transformer outward under both strategies, but far more steeply, and below the statutory limit, under uncoordinated charging.

5.6 Numerical summary

Quantity	Uncoordinated	Coordinated	Improvement
Number of houses on feeder	5	5	-
Number of EVs connected	3	3	-
Peak transformer loading (%)	69.6	36.9	32.7 % lower
Transformer overloaded (>100%)?	No	No	-
Minimum voltage at last house (V)	216.5	222.2	5.7 V higher
Maximum voltage drop (%)	5.87	3.40	2.47 % lower
Within 5% statutory voltage limit?	NO (violation)	YES	Violation resolved
Total daily energy served (kWh)	143.6	143.6	identical

Discussion

The results in Section 5 support the central hypothesis of this project directly. Although the transformer itself does not exceed its rating under either strategy at this EV penetration level (69.6% and 36.9% are both below 100%), the voltage constraint is the binding limit: uncoordinated charging produces a genuine statutory voltage violation (216.5 V, against a limit of 218.5 V) that persists for several consecutive evening hours, while coordinated charging keeps the feeder within limits at every hour of the day.

This is a physically intuitive but practically important result. Because the total energy delivered is identical in both cases (143.6 kWh, matching to within rounding in Section 5.6), the entire improvement comes from shifting when that energy is delivered, not how much is delivered. This is precisely the mechanism exploited by smart charging and time-of-use tariff schemes used by utilities to defer network reinforcement: rather than spending capital to raise the transformer rating or replace conductors with lower resistance cable to accommodate a worst case coincident peak that occurs for only a few hours per day, the utility (or an automated charging controller) can achieve the same outcome by influencing when customers charge.

The voltage profile in Figure 4 also illustrates a second useful point: the benefit of coordination is not uniform along the feeder, it grows toward the far end. House 1, closest to the transformer, sees a comparatively small difference between strategies, while House 5 sees the largest improvement. This matches the intuition that customers furthest from the source are the ones most exposed to aggregate demand effects, and are therefore the ones who benefit most from demand side coordination.

Conclusion

This project developed and validated a MATLAB GUI application that models the impact of residential EV charging on a small low voltage distribution feeder, and used it to compare uncoordinated and coordinated charging strategies under a representative EV penetration scenario. The results show that uncoordinated charging of three EVs on a five house feeder produces a genuine statutory voltage violation at the end of the feeder (5.87% drop against a 5% limit) even though the supplying transformer remains within its rated capacity, and that simply staggering the same charging events into the overnight off-peak period removes the violation entirely (reducing the drop to 3.40%) while delivering exactly the same energy to exactly the same vehicles. This confirms that, at least for the scenario studied, the network stress introduced by residential EV adoption is primarily a scheduling and control problem rather than a physical capacity problem, reinforcing the practical value of smart charging and demand side management as a lower cost alternative to network reinforcement.

Limitations and Future Work

- Resistive, unity power factor approximation: the model neglects line reactance and reactive power flow. A full AC load flow (e.g. using the backward-forward sweep method common for radial distribution feeders) would give a more precise voltage estimate, particularly for feeders with a significant X/R ratio.
- Balanced single phase representation: real LV feeders are multi-phase and frequently unbalanced, and single phase EV chargers connected to different phases would produce uneven loading across phases that this model cannot capture.
- Constant power charging profile: real EV chargers taper their charging current as the battery approaches full state of charge, following a roughly constant-current, constant-voltage profile rather than the constant power block assumed here.
- Deterministic arrival and duration: charging start times and durations are fixed by the user rather than drawn from a realistic statistical distribution of driver behaviour, journey length and battery state of charge on arrival.
- Fixed stagger interval: the coordinated strategy uses a simple, fixed interval between EV start times. A more advanced scheme would use an optimisation routine (for example, minimising peak loading or total network losses subject to each vehicle receiving its required energy by a deadline) rather than a fixed heuristic.
- Single feeder topology: the model represents one radial string of houses. A meshed or branched network, or multiple feeders sharing a transformer, would require extending the voltage drop calculation to a general network solver.

Addressing any of these points would be a natural extension of this project for future coursework or thesis-level work, particularly the introduction of a proper AC load flow solver and a stochastic, optimisation-based coordinated charging scheme.

How to Run the Application

1. Place EV_Grid_Impact_App.m in a folder and set it as the MATLAB current folder.
2. In the MATLAB Command Window, type: EV_Grid_Impact_App
3. On Tab 2 (System Setup), set the desired feeder parameters and tick the houses that own an EV.
4. On Tab 3 (Simulation), press RUN SIMULATION, then press ANIMATE 24 HOURS to view the hour-by-hour build-up of the evening peak.
5. On Tab 4 (Results), review the numerical comparison table, the written verdict, and optionally press Export Results to CSV.